

Code-Layout, Readability and Reusability

Date	2 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy
Maximum Marks	

Code-Layout, Readability and Reusability:

This document evaluates the quality of the FitBuddy codebase in terms of its structure, readability, and reusability. FitBuddy is built with a FastAPI backend, Jinja2 templates, and Gemini AI integration. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation in Python and HTML templates
2	Proper File Structure	Files and folders are logically organized	Yes	Separate routes, templates and static files
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names for user inputs and AI responses
4	Function / Method Names	Functions are descriptively named	Yes	Route and helper functions named by purpose
5	Code Comments	Inline and block comments explain logic	Partial	Key logic commented; can be extended
6	Modular Design	Code is split into reusable functions/modules	Yes	Gemini logic kept separate from routing
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared base template avoids repetition
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Input validation and API error handling

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini AI Integration Module	Python, Gemini API	Fitness responses across app features	High
2	FastAPI Routing and Input Handling	Python, FastAPI	All user-facing pages and form submissions	High
3	Base Jinja2 Template	HTML, Jinja2	All dynamic pages	High
4	Frontend UI Components	HTML, CSS	Forms, result and navigation sections	Medium

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
5	Frontend-Backend Connector	FastAPI, Jinja2	Rendering data into templates	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of backend, templates and AI logic
Readability	4	Consistent indentation and descriptive naming
Reusability	4	Modular Gemini, routing and template components
Documentation / Comments	3	Core logic commented; more docstrings can be added
Overall Score	4	Well-structured and maintainable codebase